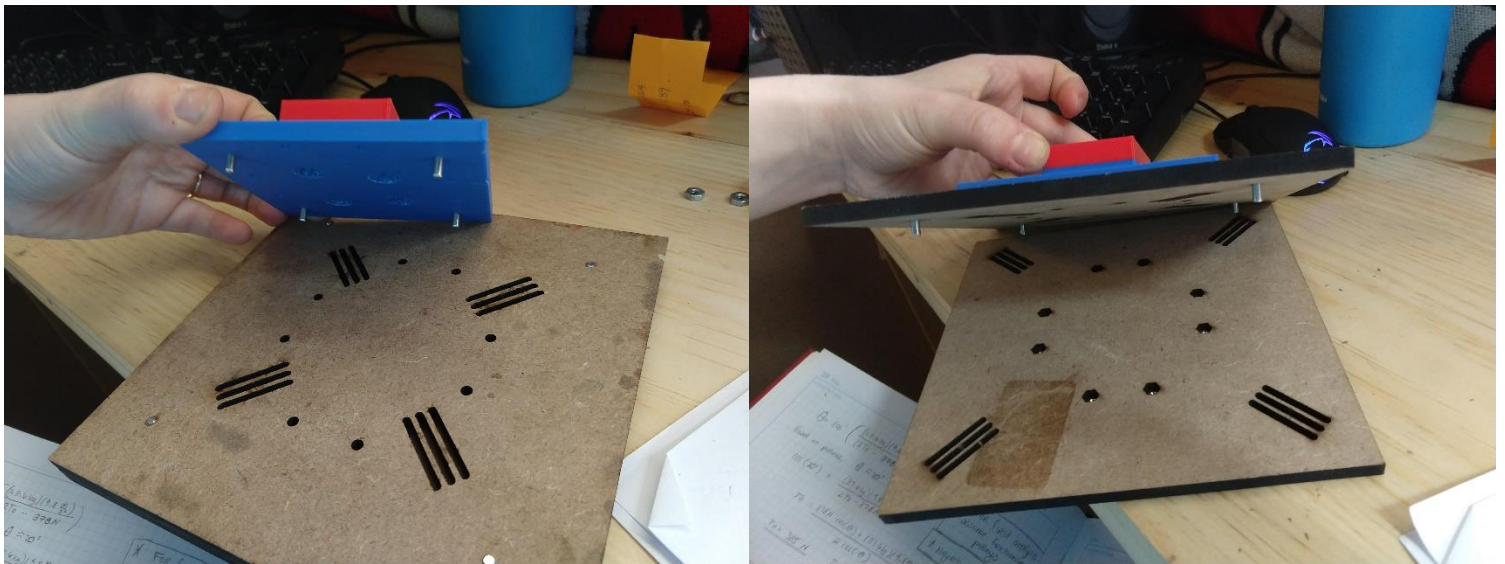
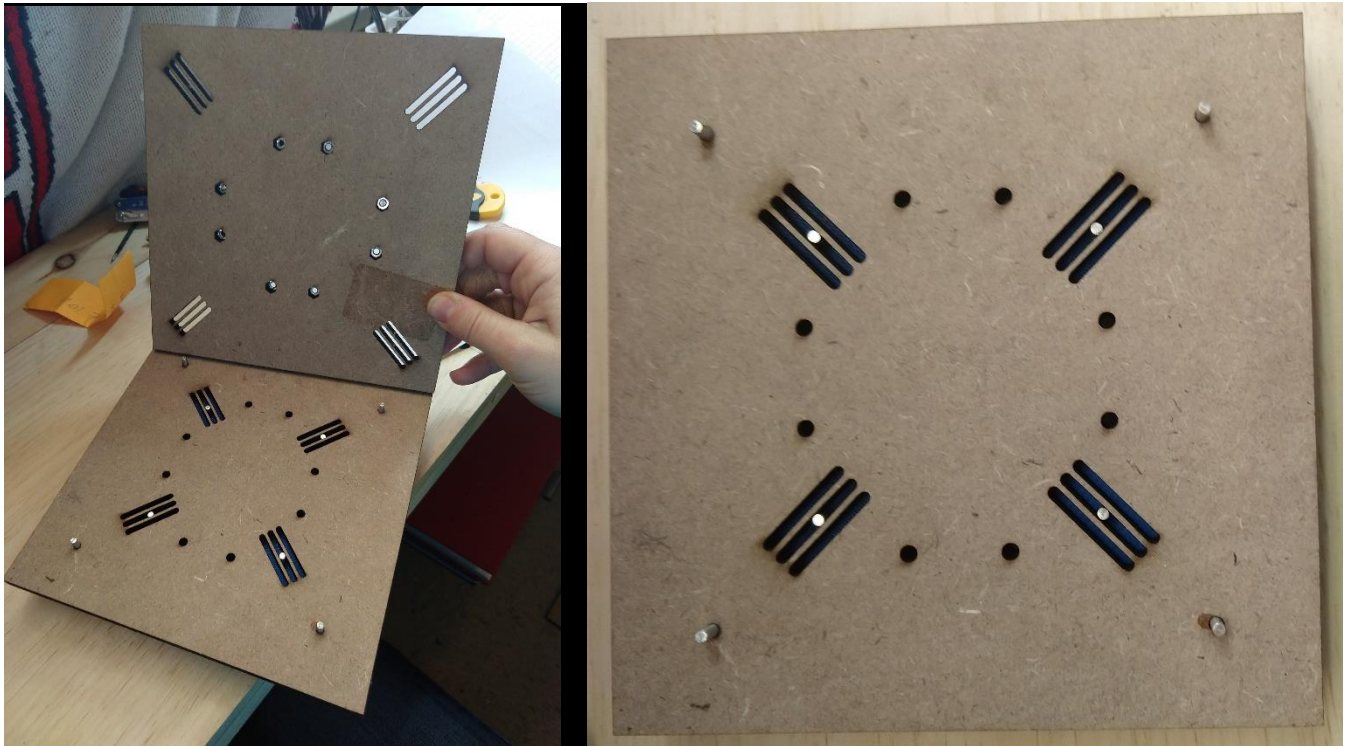
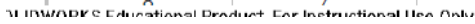
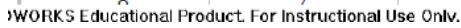


APPENDIX









Rachel Hoffman
Elastically Averaged Coupling Assignment
2.77 2/22/2018

```
%Elastic Averaging Calculations
%Rachel Hoffman 2/20/2018
%https://ac.els-cdn.com/S014163591630407X/1-s2.0-S014163591630407X-
main.pdf?_tid=9c3fb4ba-1691-11e8-b9d4-
00000aab0f26&acdnat=1519167708_f15e09eedf572f5bac4c5a12399a1ad0
% • Define the desired accuracy of the assembly (da).
% • Define the expected geometric mismatch of the parts (manufacturing
% error) (dm). This is also the overall amount that elastic
% contacts can be expected to have to deform to accommodate
% manufacturing errors.
% • The factor Ns to achieve better performance is in a ratio;
% Ns = dm/da.
% • Assuming the manufacturing errors will be randomly distributed,
% the number of elastic contacts, N, required to "average out" the
% errors will be Ns = N^2 conservatively.
% • Define the desired stiffness of the assembled joint, ka.
% • Set the number of elastic contacts.
%measured 3.75mm, .148in -.125+ .023in left over

%Deviations along x,y,theta directions
%[dX; dY ;thetaZ] = inv[Ksys] [Fx; Fy; Mz]

%KiE=KiTangential+KiNormal, i denotes numbering of each dowel.
%KiN must be oriented to kiT before summation

%-----
%-----
%Properties of MDF Flexure
%USER CAN CHANGE THESE
E=4*10^9;
YieldStress = 18*10^6; %UTS of MDF
b=.25*.0254; %thickness of MDF (1/4")
l=1.5*.0254;%.0254; %1inch
t=.08*.0254;%.0254;%.0735*.0254 %width of flexure, variable parameter, Lasercutter Error width
measured+.002in;
A=b*t;
L=l/2;
r=(3)*.0254;% coupling radius
%Properties of DowelPin
Dpin=(1/8)*.0254;%=.1875 in;
Dpinmax=Dpin;
Dpinmin=Dpin;
%slotgap=.115
DelDes=Dpin-(.115*.0254)/2 *.0254; %DelDes=50*10^-6;%(Dpin-.115*.0254)/2 *.0254; %50*10^-6
AL recommended from paper
%StiffnessOnecantilever= F/(F*L^3/(3*E*I))
%Accuracy of LaserCutter
LcutAcc= .254/1000;%
LcutRep=0.0127/1000;% meters

%ERROR OFFSET OF PIN LOCATION
%pin1,pin2,pin3,pin4 Radial offset. meters
OFFSET=[.000;0;0;0];
maxOffset= 2*LcutAcc + DelDes; % Max possible offset the flexure will experience due to pin location
and initial displacement
%*2LcutAcc because acc of slot and acc of pin location
%Don't Change These
V=Dpinmax/2 - DelDes + t/2;
%W=(Dpinmax/2)*sin(acos(((Dpinmax/2)-DelDes)/(Dpinmax/2)));
W=sqrt(Dpin^2 -(Dpin-DelDes)^2)
%Spacing between centers
%of beam flexures less than pin diameter

%Sanity Check
FlexureStressNom= 3*E*DelDes*(t/2)/L^2
StressRatioNom=YieldStress/FlexureStressNom %should be >1
maxdeflectionbeforeyield_mm= ((YieldStress*L^2)/(3*E*(t/2)))*1000
maxdeflectionbeforeyield_inch= maxdeflectionbeforeyield_mm/25.4

FlexureStressMax= 3*E*maxOffset*(t/2)/L^2
StressRatioMax=YieldStress/FlexureStressMax %should be >1
%%
%-----
%-----

l=(1/12)*b*t^3;
a1=E*A/2*L;
a2=12*E*I/L^3;
a3=6*E*I/L^2;
a4=4*E*I/L;

kbj1=[a1 0 0; 0 a2 -a3; 0 -a3 a4];
kbj2=[a1 0 0; 0 a2 a3; 0 a3 a4];
kbj3=kbj1;
kbj4=kbj2;
%half the distance between the center of the two beam
%elements, 2V, and half the distance of the contact surface between
%the dowel pin and the pair of flexures, 2W.
ry1=V;
```

```
ry2=V;
ry3=-V;
ry4=-V;
rx1=W;
rx3=W;
rx2=-W;
rx4=-W;
Jj1=[ 1 0 -ry1
      0 1 rx1
      0 0 1];
Jj2=[ 1 0 -ry2
      0 1 rx2
      0 0 1];
Jj3=[ 1 0 -ry3
      0 1 rx3
      0 0 1];
Jj4=[ 1 0 -ry4
      0 1 rx4
      0 0 1];
%Spring stiffness of BEAM ELEMENTS with respect to its own coordinate system
k1= inv(Jj1)*kbj1*inv(Jj1) ;
k2= inv(Jj2)*kbj2*inv(Jj2);
k3= inv(Jj3)*kbj3*inv(Jj3);
k4= inv(Jj4)*kbj4*inv(Jj4);

%Spring stiffness of tangential spring
%(All springs have same geometry)
K11=k1+k2+k3+k4;
K12=k1+k2+k3+k4;
K13=k1+k2+k3+k4;
K14=k1+k2+k3+k4;
theta1=pi/4;
theta2=theta1+pi/2;
theta3=theta2+pi/2;
theta4=theta3+pi/2;
%spring stiffness with respect to Top Plate Coordinate System
G1=[cos(theta1) cos(theta1+pi/2) 0
     sin(theta1) sin(theta1 +pi/2) 0
     0 0 1];
G2=[cos(theta2) cos(theta2+pi/2) 0
     sin(theta2) sin(theta2 +pi/2) 0
     0 0 1];
G3=[cos(theta3) cos(theta3+pi/2) 0
     sin(theta3) sin(theta3 +pi/2 ) 0
     0 0 1];
G4=[cos(theta4) cos(theta4+pi/2) 0
     sin(theta4) sin(theta4 +pi/2) 0
     0 0 1];
ry1=0;
ry2=0;
ry3=0;
ry4=0;
rx1=r;
rx2=r;
rx3=r;
rx4=r;
Ji1=[ 1 0 -ry1
      0 1 rx1
      0 0 1];
Ji2=[ 1 0 -ry2
      0 1 rx2
      0 0 1];
Ji3=[ 1 0 -ry3
      0 1 rx3
      0 0 1];
Ji4=[ 1 0 -ry4
      0 1 rx4
      0 0 1];

KOp1=G1*inv(Ji1)*K11*inv(Ji1)*G1';
KOp2=G2*inv(Ji2)*K12*inv(Ji2)*G2';
KOp3=G3*inv(Ji3)*K13*inv(Ji3)*G3';
KOp4=G4*inv(Ji4)*K14*inv(Ji4)*G4';

%Stiffness of each segment w.r.t O' (Body)
wbod=b;
tbod=2*pi*(r+l)/4;
Lbod=r-L;
Abod=tbod*wbod;
Ebod=E;
lbod=(1/12)*tbod^3*wbod;
k1B=[Ebod*Abod/Lbod 0 0; 0 12*Ebod*lbod/Lbod^3 -6*Ebod*lbod/Lbod^2; 0 -6*Ebod*lbod/Lbod^2
4*E*lbod/Lbod];
k2B=k1B;
k3B=k1B;
k4B=k1B;
%
KOpB1=G1*k1B*G1';
KOpB2=G2*k2B*G2';
KOpB3=G3*k3B*G3';
KOpB4=G4*k4B*G4';

%
Ksys = inv(inv(KOp1) + inv(KOpB1)) + inv(inv(KOp2) + inv(KOpB2)) +inv(inv(KOp3) + inv(KOpB3))
+inv(inv(KOp4) + inv(KOpB4) );
%
K22=192*E*I/l^3;
```

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```
%Offset of each Dowel Pin Radially
CM=[1 0.5 0 0.5; 0.5 1 0.5 0; 0 0.5 1 0.5; 0.5 0 0.5 1]

DELTA= CM*OFFSET;
del1=DELTA(1);
del2=DELTA(2);
del3=DELTA(3);
del4=DELTA(4);

gs=(2*v-b)/2;
go=Dpinmax/2;

Fa1= K22*[(gs+OFFSET(1))-go];
Fb1= K22*[(gs-OFFSET(1))-go];

Fa2= K22*[(gs+OFFSET(2))-go];
Fb2= K22*[(gs-OFFSET(2))-go];

Fa3= K22*[(gs+OFFSET(3))-go];
Fb3= K22*[(gs-OFFSET(3))-go];

Fa4= K22*[(gs+OFFSET(4))-go];
Fb4= K22*[(gs-OFFSET(4))-go];

%Use Fa,Fb equs with epsi=deli to determine each
%pair of forces from each elastic contact

Fxp1= ((Fb1-Fa1)*sin(theta1))+((Fb2-Fa2)*sin(theta2))+((Fb3-Fa3)*sin(theta3))+((Fb4-
Fa4)*sin(theta4));
Fyp1= ((Fb1-Fa1)*cos(theta1))+((Fb2-Fa2)*cos(theta2))+((Fb3-Fa3)*cos(theta3))+((Fb4-
Fa4)*cos(theta4));
Mzp1=((Fb1-Fa1)*r)+((Fb2-Fa2)*r)+((Fb3-Fa3)*r)+((Fb4-Fa4)*r);

Q=ones(4);
e=sum(OFFSET)/4;
Ahat=e*ones(4,1);

CMp=CM/4;

EquipPos=CMp*OFFSET;

R=(Q-CM)*(Ahat-EquipPos);

v1=R(1);
v2=R(2);
v3=R(3);
v4=R(4);

Fa1r= K22*[(gs+v1)-go];
Fb1r= K22*[(gs-v1)-go];

Fa2r= K22*[(gs+v2)-go];
Fb2r= K22*[(gs-v2)-go];

Fa3r= K22*[(gs+v3)-go];
Fb3r= K22*[(gs-v3)-go];

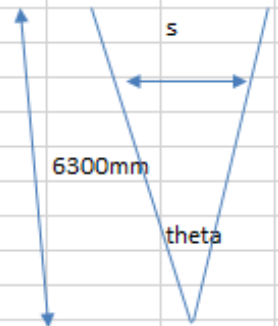
Fa4r= K22*[(gs+v4)-go];
Fb4r= K22*[(gs-v4)-go];

Fxp2= ((Fb1r-Fa1r)*sin(theta1))+((Fb2r-Fa2r)*sin(theta2))+((Fb3r-Fa3r)*sin(theta3))+((Fb4r-
Fa4r)*sin(theta4));
Fyp2= ((Fb1r-Fa1r)*cos(theta1))+((Fb2r-Fa2r)*cos(theta2))+((Fb3r-Fa3r)*cos(theta3))+((Fb4r-
Fa4r)*cos(theta4));
Mzp2=((Fb1r-Fa1r)*r)+((Fb2r-Fa2r)*r)+((Fb3r-Fa3r)*r)+((Fb4r-Fa4r)*r);

Fx= Fxp1 + Fxp2;
Fy=Fyp1+Fyp2;
Mz=Mzp1+Mzp2;

POSCCHANGE=inv(Ksys)*[Fx;Fy;Mz]
Ksys
Ksysum=Ksys*10^6
50/Ksysum(1)
```

Applied Torque Nm	1.524			
Trial	Theta-deflection (mm) projected abbe	Theta-Deflection with applied load		
1	5.87	15.93		
2	7.44	16.25		
3	7.73	16.84		
4	6.96	16.43		
5	8.04	17.37		
6	8.8	17.45		
7	8.25	18.17		
8	9.05	17.86		
9	8.75	18.38		
10	8.44	18.2		
Average	7.933	17.288		
arc length deflection	9.355			
theta=	0.001484921			
stiffness N/rad	1026.317477			
Stiffness N/microrad	0.001026317			
Predicted from MATLAB	0.003			
%diff	65.78941742			



A	B	C	D	E	F
Applied Force, N =	40				
	One 'tick' on dial indicator = .0005"				
Trial	y-deflection ('ticks')	x-deflection ('ticks')			
1	12	11.5			
2	13.5	13			
3	14	13.5			
4	13.5	13.5			
5	12.5	12			
6	12.5	14			
7	13	13.5			
8	14.5	14.5			
9	13	14			
10	13.5	11.5			
	y-deflection , um	x-deflection, um			
1	152.4	146.05			
2	171.45	165.1			
3	177.8	171.45			
4	171.45	171.45			
5	158.75	152.4			
6	158.75	177.8			
7	165.1	171.45			
8	184.15	184.15			
9	165.1	177.8			
10	171.45	146.05			
Average	167.64	166.37			
Measured Stiffness N/um	0.238606538	0.240427962			
Predicted Stiffness N/um from EACCalculations_RMH_Rev2.m	0.2502	0.2502			
	4.63367793	3.905690739			

A	B	C	D	E	F	G
NumcontactPOints	8					
Manufacturing Error (um)	12.7					
	One 'tick' on dial indicator = .0005"					
Trial	x-deflection (inches)	y-deflection (inches)	Theta-deflection (mm) projected abbe			
1	0.00025	0.00025	8			
2	0.00055	0.0005	7.44			
3	0.0005	0.0005	7.73			
4	0.0005	0.00075	6.96			
5	0	0.00025	8.04			
6	0.0005	0.0005	8.8			
7	0.00025	0.00025	8.25			
8	0.00025	0.00025	9.05			
9	0.00055	0.0005	8.75			
10	0.00025	0.0005	8.44			
	x-deflection, um	y-deflection, um	theta-deflection, um projected abbe			
1	6.35	6.35	8000			
2	13.97	12.7	7440			
3	12.7	12.7	7730			
4	12.7	19.05	6960			
5	0	6.35	8040			
6	12.7	12.7	8800			
7	6.35	6.35	8250			
8	6.35	6.35	9050			
9	13.97	12.7	8750			
10	6.35	12.7	8440			
Average	0	0	8.146	Avg mm from reference		
StdDev	4.706431061	4.285923341	0.0001034	theta std dev, rad		
2 Standard Deviations	9.412862122	8.571846683	0.0002068			
		theta, mrad	0.1033798			
		theta, deg	0.0059232			
		651.2927316	0.0001034			
Deviations along x and y axis estimated (err/sqrt(numpts))	4.490128061	4.490128061				
Measured StdDev	4.706431061	4.285923341				
Percent Difference	4.817301371	4.547859578				
rotational repeatability predicted, microrad	6.35					
rotational repeatability measured, microrad	6.365503297					
Theta	0.083536789					
ThetaPredicted	0.083333333					
P						

Comparing manufactured dimensions with solidworks dimensions			
Trial	length of flexure,mm	width of flexure,mm	thickness of assembly,mm
1	28.18	1.62	12.93
2	28.23	1.62	12.92
3	28.26	1.63	12.89
4	28.32	1.51	12.93
5	28.25	1.65	12.79
6	28.31	1.64	12.86
7	28.28	1.66	12.91
8	28.19	1.65	12.81
9	28.24	1.66	12.88
10	28.27	1.64	12.86
Average,mm	28.253	1.628	12.878
Standard Deviation ,mm	0.045716518	0.043919118	0.048716869
Solidworks Dimensions, mm	23.548	1.592	12.7
Percent Difference	19.98046543	2.261306533	1.401574803

Stress	
flexure deflection desired	0.000127
manufacturing error	0.0000127
max deflection	0.0001397
UTS Abs	18000000
E	4000000000
thickness	0.002032
max width	0.00667
width	0.00635
Area	1.29032E-05
Max Area	1.35484E-05
I	0.0127
Moment of Inertia	4.4398E-12
Max Moment of Inertia	4.66179E-12
Force ,N	3.63354112
Max Force	3.815218176
Bending Stress	10560000
MaxBendingStress	10560000
Will it Break?	No